

CSE220 Lab3: Final Project

Final Due Date: June 11, 2026

Project Presentation: Last week of class

In this lab, you will implement a microarchitectural technique in Scarab of your choice. There is no correct or incorrect submission for this lab. We will evaluate the 1) correctness of your proposed architecture, 2) correctness of your implementation, and 3) performance analysis and evaluation of your technique. For instance, it is ok if your technique does not provide performance gains compared to the baseline as long as you can explain why this is the case. For instance, if you are implementing a branch predictor, we do not expect you to beat TAGE; instead, you want to compare it to a simple bimodal predictor. You also do not need to implement every aspect of a technique presented in the paper. Start with a basic design and add features iteratively. You need to determine a good baseline and evaluation methodology. Below are listed some potential paper candidates for implementation. You may choose one of these candidates or propose your own. This lab has three submissions.

1. Project Plan: This should be a very short description of the project plan. It should briefly describe the technique you are planning to implement. It should explain how the technique can be integrated into scarab. For instance, it should describe which input data your technique requires (e.g. L1 accesses) and what outputs it generates. The plan should also list the source code files you will need to modify and a rough sketch of the implementation e.g. what data structures it will use.
2. Project Check-In: Submit a git repository link referencing the current source-code implementation within Scarab.
3. Final Report: This should be ~2 pages in length. It should describe the implemented technique and how it was integrated into Scarab. It should also describe your evaluation methodology and results (similar to previous labs).
4. Project Presentation: We will use the last week of class for short presentations (10 min for each group).

Note: This lab is intentionally designed to be open-ended, resembling the type of work a computer architect performs. Try to come up with solutions to the challenges you are facing. If you need help, ask the course team.

Paper implementation candidates:

1. Daniel Jimenez, Calvin Lin: Dynamic branch prediction with perceptrons
2. Yeh, Tse-Yu, and Yale N. Patt. "Two-level adaptive training branch prediction." Proceedings of the 24th annual international symposium on Microarchitecture. 1991.
3. J.Kim, S.H.Pugsley, P.V.Gratz, A.Reddy, C.Wilkerson ,and Z.Chishti, "Path confidence based lookahead prefetching," in MICRO, 2016.
4. P.Michaud, "Best-offset hardware prefetching," in HPCA, 2016.
5. S. Somogyi, T. F. Wenisch, A. Ailamaki, B. Falsafi, and A. Moshovos, "Spatial memory streaming," in ISCA, 2006
6. Mohammad Bakhshalipour, Mehran Shakerinava, Pejman Lotfi-Kamran, and Hamid Sarbazi-Azad. "Bingo Spatial Data Prefetcher." In HPCA. 2019.
7. Mohammad Bakhshalipour, Pejman Lotfi-Kamran, and Hamid Sarbazi-Azad. "Domino Temporal Data Prefetcher." In HPCA. 2018
8. SWQUE: A Mode Switching Issue Queue with Priority-Correcting Circular Queue, Hideki Ando (Nagoya University)
9. Yasuo Ishii, Mary Inaba, and Kei Hiraki "Access map pattern matching for data cache prefetch" ICS '09
10. J. Bonanno, A. Collura, D. Lipetz, U. Mayer, B. Prasky, and A. Saporito, "Two Level Bulk Preload Branch Prediction," in International Symposium on High-Performance Computer Architecture, pp. 71–82, 2013
11. M. Kharbutli and Y. Solihin, "Counter-based cache replacement and bypassing algorithms," IEEE Trans. Comput., vol. 57, no. 4, pp. 433–447, Feb. 2008.
12. Nam Duong, Dali Zhao, Taesu Kim, Rosario Cammarota, Mateo Valero, and Alexander V. Veidenbaum. 2012. Improving Cache

Management Policies Using Dynamic Reuse Distances. In 45th International Symposium on Microarchitecture (MICRO). 389–400.